Predicate Logic

Not all valid deductions can be formalised in propositional logic.

Example: To express the following logical deduction in propositional logic:

1. All horses are animals.
2. *Therefore,* all horse-heads are animal heads.

we would need to say things like *"if* x *is a horse, then* x *is an animal"*.

In order to do this, we extend propositional logic to include ***predicates***: properties which are true or false of elements of a universe of discourse.

We have actually been using predicates with set-building notation

$$\{\, \mathrm{x} : \mathrm{x} \textit{ has property } \mathrm{P} \,\}$$

The part "x *has property* P" is a predicate, typically denoted $\mathrm{P}(\mathrm{x})$. A predicate is an *indeterminate proposition*, which may be true or false of a particular element.

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Predicates and Free Variables

Example:

$$\text{Prime}(x) \quad = \quad \text{“}x \text{ is a prime number”}$$

is a predicate which says that x is a prime number. The universe of discourse is probably $\mathbb{N}$; but it could be anything (and $\text{Prime}(x)$ would be false if $x \notin \mathbb{N}$).

The “x” in “$\text{Prime}(x)$” is called a ***free variable*** as it stands for an unspecified value. If we instantiate x, we would get a proposition. For example

$$\text{Prime}(5) \text{ is true} \quad \text{whereas} \quad \text{Prime}(6) \text{ is false}$$

Example: Given the predicate $\text{Female}(x) = \text{“}x \text{ is a female”}$ where the universe of discourse is the Duck family:

$$\text{Ducks} = \{ \text{Quackmore, Hortense, Scrooge, Donald, Della, Huey, Louis, Dewey} \}$$

Then

- $\text{Female}(\text{Hortense})$ and $\text{Female}(\text{Della})$ are true;
- $\text{Female}(\text{Quackmore})$, $\text{Female}(\text{Scrooge})$, $\text{Female}(\text{Donald})$, $\text{Female}(\text{Huey})$, $\text{Female}(\text{Louis})$ and $\text{Female}(\text{Dewey})$ are false.

Truth Sets

The ***truth set*** of a predicate $P(x)$ is the set of objects which satisfy it:

$$\{\, x : \; P(x) \,\}$$

Examples:

The truth set of $\mathrm{Prime}(x)$ is the set of all prime numbers:

$$\{\, 2\,,\, 3\,,\, 5\,,\, 7\,,\, 11\,,\, 13\,,\, 17\,,\, \ldots \,\}.$$

The truth set of the predicate $\mathrm{Female}(x)$ over the Duck family is

$$\{\, \mathrm{Hortense},\ \mathrm{Della} \,\}.$$

The truth set of the predicate *"x is an even integer"* is

$$\{\, \ldots\,,\, -6\,,\, -4\,,\, -2\,,\, 0\,,\, 2\,,\, 4\,,\, 6\,,\, \ldots \,\}.$$

The truth set of the predicate *"x is an even prime number"* is $\{\, 2 \,\}$.

Predicates with two or more free variables

Predicates may range over more than one element.

For example, *equality* ($=$) and *set inclusion* ($\subseteq$) are binary predicates. For these, we typically use ***infix notation*** instead of ***prefix notation***; that is, $x{=}y$ instead of $=(x, y)$, and $A{\subseteq}B$ instead of $\subseteq(A, B)$.

For example: $5=5$ is true; $\emptyset \subseteq \{5\}$ is true; and $\{\emptyset\} = \emptyset$ is false.

The truth set of a predicate which ranges over more than one element consists of a tuple of values.

Example: Let $\text{Divides}\,(x, y)$ be the two-place predicate over integers saying that x *divides evenly into* y. Then

- $\text{Divides}\,(3, 15)$ is true, and $\text{Divides}\,(4, 15)$ is false;
- the truth set of $\text{Divides}\,(x, y)$ is

$$\{\,(x, y) :\ x \text{ divides evenly into } y\,\}$$

(Note: The standard mathematical symbol for Divides is "$|$", used in infix notation, as in $3 \mid 15$ and $4 \nmid 15$.)

All and Some

We can form a proposition from a (unary) predicate by saying that it is true of *every* element, or that it is true of *some* (i.e., at least one) element.

Example: *"Every prime number greater than 2 is odd"*:

$$\mathrm{Odd}(3) \wedge \mathrm{Odd}(5) \wedge \mathrm{Odd}(7) \wedge \mathrm{Odd}(11) \wedge \cdots$$

where $\mathrm{Odd}(x)$ says *"x is an odd number"*.

Example: *"Some prime numbers are square"*:

$$\mathrm{Square}(2) \vee \mathrm{Square}(3) \vee \mathrm{Square}(5) \vee \cdots$$

where $\mathrm{Square}(x)$ says *"x is a square number"*.

These are infinitely-long conjunctions / disjunctions, and cannot be expressed in propositional logic (without using an ellipsis $\cdots$).

Predicate logic provides two forms of *quantification* which will allow to express these in a formula in a finite way: one which expresses that a property holds univerally (always), and another which expresses that a property holds existentially (sometimes).

Universal Quantification

The statement $\forall x\ P(x)$ (pronounced *"for all x, $P(x)$"*) is called ***universal quantification*** and is true if, and only if, $P(x)$ is true of ***all*** possible values of x.

Example: *"Everyone did the homework"* is expressed as:

$$\forall x\, H(x)$$

where $H(x) =$ "x did the homework", and the (assumed) universe of discourse is the set of students who were assigned the homework.

- the variable x in the formula $\forall x\, H(x)$ is called a ***bound variable***; it is *bound* by the quantifier $\forall x$.
- The statement has a definite truth value.

Example: *"Every prime number greater than 2 is odd"* is expressed as:

$$\forall x \Big(\text{Prime}(x) \ \wedge\ x > 2 \ \Rightarrow\ \text{Odd}(x) \Big)$$

Note: The (outer) parentheses are needed, as universal quantification binds more strongly than all propositional connectives.

Universal Quantification

Example: *"Every dog that has stayed in the kennel will have to go into quarantine"* can be expressed as

$$\forall x \left(K(x) \Rightarrow Q(x) \right)$$

where $K(x) =$ "x *has stayed in the kennel*" and $Q(x) =$ "x *has to go into quarantine*". The (assumed) universe of discourse is the set of all dogs.

Breaking this down:

For all x: if x has stayed in the kennel, then x has to go in to quarantine.

Example: *"Nobody likes a sore loser"* can be expressed as:

$$\forall x \left(S(x) \Rightarrow \forall y \neg L(y, x) \right)$$

where $S(x) =$ "x *is a sore loser*" and $L(y, x) =$ "y *likes* x". The (assumed) universe of discourse is set of all people.

Breaking this down:

For all x: if x is a sore loser, then for all y: y doesn't like x.

Universal Quantification

Exercise: Using the predicates

- $B(x)$ = "x *is a bee*"
- $F(x)$ = "x *is a flower*"
- $L(x, y)$ = "x *likes* y"

write each of the following statements in predicate logic.

1. Bees like all flowers *(For all x: if x is a bee, then for all y: if y is a flower, then x likes y)*

$\forall$ x (B(x) $\Rightarrow$ $\forall$y (F(y) $\Rightarrow$ L(x,y)))

$\forall$ x $\forall$y (B(x) $\Rightarrow$ F(y) $\Rightarrow$ L(x,y))

$\forall$ x $\forall$y (B(x) $\wedge$ F(y) $\Rightarrow$ L(x,y))

2. Bees only like flowers *(For all x: if x is a bee, then for all y: if x likes y, then y is a flower)*

$\forall$ x (B(x) $\Rightarrow$ $\forall$y (L(x,y) $\Rightarrow$ F(y)))

3. Only bees like flowers *(For all x and for all y: if y is a flower and x likes y then x is a bee)*

$\forall$ x $\forall$ y (L(x,y) $\wedge$ F(y) $\Rightarrow$ B(x))

$\forall$ x (($\exists$ y L(x,y) $\wedge$ F(y)) $\Rightarrow$ B(x))

Existential Quantification

The statement $\exists x\ P(x)$ (pronounced *"there exists* x *such that* $P(x)$*"*) is called ***existential quantification*** and is true if, and only if, $P(x)$ is true of ***some*** (at least one) value of x.

Example: *"Some prime numbers are square"* can be expressed as:

$$\exists x\Big(\text{Prime}(x) \wedge \text{Square}(x)\Big)$$

- the variable x in the formula is (again) called a ***bound variable***; it is bound by the quantifier $\exists x$.
- The statement has a definite truth value (it is false).

Example: *"Someone didn't do the homework"* is expressed as:

$$\exists x\ \neg\, H(x)$$

where $H(x) =$ "x *did the homework*", and the (assumed) universe of discourse is the set of students who were assigned to do the homework.

Note: Existential quantification also binds more strongly than all propositional connectives.

Existential Quantification

Example: *"If some dog that has stayed in the kennel has been in contact with a dog with rabies, then every dog that has stayed in the kennel will have to go into quarantine"* can be expressed as

$$\exists x \Big(\mathrm{K}(x) \wedge \exists y \Big(\mathrm{C}(x,y) \wedge \mathrm{R}(y) \Big) \Big) \Rightarrow \forall x \Big(\mathrm{K}(x) \Rightarrow \mathrm{Q}(x) \Big)$$

where
$$\begin{aligned} \mathrm{K}(x) &= \text{"}x \textit{ has stayed in the kennel}\text{"} \\ \mathrm{R}(x) &= \text{"}x \textit{ has rabies}\text{"} \\ \mathrm{C}(x,y) &= \text{"}x \textit{ and } y \textit{ have been in contact}\text{"} \\ \mathrm{Q}(x) &= \text{"}x \textit{ has to go into quarantine}\text{"} \end{aligned}$$

The (assumed) universe of discourse is the set of all dogs.

Breaking this down:

If there is an x that has stayed in the kennel,
and there is a y that has been in contact with x and has rabies,
then for all x: if x has stayed in the kennel,
then x has to go into quarantine.

Existential Quantification

Exercise: Let $M(x, y)$ say *"x is the mother of y"*, where the universe of discourse is the set of all human beings.



Express the following predicates

1. Everyone has a mother

   (For each x *there is a* y *such that*
   y *is the mother of* x*)*

   $\forall$ x $\exists$ y M(y,x)

2. Everyone has exactly one mother

   (For each x *there is a* y *such that*
   y *is the mother of* x*, and*
   for every z*: if* z *is the mother of* x*,*
   then $z = y$*)*

$\forall$ x $\exists$ y (M(y,x) $\wedge$ $\forall$ z (M(z,x) $\Rightarrow$ z = y))

Bounded Quantification

The following are useful restricted forms of quantification which we use for convenience:

- $\forall x \in A\, P(x)$ pronounced *"for all x in A, $P(x)$"*

  is logically equivalent to: $\forall x \left(x \in A \Rightarrow P(x) \right)$

- $\exists x \in A\, P(x)$ pronounced *"there exists some x in A such that $P(x)$"*

  is logically equivalent to: $\exists x \left(x \in A \wedge P(x) \right)$

- $\exists! x\, P(x)$ pronounced *"there exists a unique x such that $P(x)$"*

  is logically equivalent to: $\exists x \left(P(x) \wedge \neg \exists y \left(P(y) \wedge x \neq y \right) \right)$

`∃!x ∈ A P(x)` `translates to`
`∃x ∈ A (P(x) ∧ ¬ ∃y ∈ A(P(y) ∧ x ≠ y))`

Rules for Quantification

- If it is not the case that $P(x)$ is true for all x,
  then there must be some x for which $P(x)$ is false:

$$\neg\forall x\, P(x) \;\Leftrightarrow\; \exists x\, \neg P(x)$$

- If it is not the case that $P(x)$ is true for some x,
  then $P(x)$ must be false for all x:

$$\neg\exists x\, P(x) \;\Leftrightarrow\; \forall x\, \neg P(x)$$

Examples:

¬(Some people like mathematics) ⇔ Everybody dislikes mathematics

¬(All cats have fur and a tail) ⇔ Some cats are either bald or tailless

```
¬ ∀ x ∈ Cat ( Fur(x) ∧ Tail(x) )
  ⇔ ∃ x ∈ Cat (¬Fur(x) ∨ ¬Tail(x))
```

Modelling in Predicate Logic

Exercise: Let *Divides* (k, n) be the statement "k divides evenly into n", where the variables range over the set of positive integers $\mathbb{Z}^+$; that is, $n{=}dk$ for some integer d.

Determine the truth value of each of the following statements.

1. *Divides* $(4, 6)$ **`false`**
2. *Divides* $(2, 6)$ **`true`**
3. $\forall k \, \forall n$ *Divides* (k, n) **`false`**
4. $\exists n \, \forall k$ *Divides* (k, n) **`false`**
5. $\exists k \, \forall n$ *Divides* (k, n) **`true`**
6. $\forall n \, \exists k$ *Divides* (k, n) **`true`**

```
3 is false, because 4 does not
divide 6

4 is false, because if n is
any positive number, then
n + 1 does not divide n

5 is true, because the number
k = 1 divides all integers.

6 is true, because if n is
any positive integer, then
1 is a positive integer that
divides n.
```



Last week's quiz

Question 9:

LT = "<" on $\mathbb{N}$

What is $\mathrm{LT} \circ \mathrm{LT}$?

$$\mathrm{LT} \circ \mathrm{LT} = \left\{(x, z) \in \mathbb{N}^2 \mid \exists y.(x < y \wedge y < z)\right\}$$

$$\subseteq \mathrm{LT}$$

But this inclusion is strict, because for example
x < x + 1 for all x
but there does not exist any y with
x < y < x + 1

Question 10: $\mathrm{LE} = \text{“}\leq\text{”}$

What is $\overline{\mathrm{LE}} \circ \mathrm{LE}$?

$$\overline{\mathrm{LE}} \circ \mathrm{LE} = \left\{(x, z) \in \mathbb{N}^2 \mid \exists y.\, (x \leq y \wedge y > z)\right\}$$

$$= \mathbb{N} \times \mathbb{N}$$

Modelling in Predicate Logic

Example: A Lewis Carroll puzzle deduces that no baby can manage a crocodile from the following assumptions:

$$\neg\exists x\,(B(x) \wedge M(x))$$
$$\Leftrightarrow \forall x\,(\neg B(x) \vee \neg M(x))$$
$$\Leftrightarrow \forall x\,(B(x) \Rightarrow \neg M(x))$$

- All babies are illogical.
- Nobody is despised who can manage a crocodile
- Illogical persons are despised.

Define the following predicates:

$\mathrm{B}(\mathrm{x}) =$ "x is a baby"
$\mathrm{I}(\mathrm{x}) =$ "x is illogical"
$\mathrm{D}(\mathrm{x}) =$ "x is despised"
$\mathrm{M}(\mathrm{x}) =$ "x can manage a crocodile"

The three premises above translate into the following:

1. $\forall \mathrm{x}\left(\mathrm{B}(\mathrm{x}) \Rightarrow \mathrm{I}(\mathrm{x})\right)$
2. $\forall \mathrm{x}\left(\mathrm{M}(\mathrm{x}) \Rightarrow \neg \mathrm{D}(\mathrm{x})\right)$ $\quad \forall x.\,(D(x) \Rightarrow \neg M(x))$
3. $\forall \mathrm{x}\left(\mathrm{I}(\mathrm{x}) \Rightarrow \mathrm{D}(\mathrm{x})\right)$

and the conclusion translates to $\forall \mathrm{x}\left(\mathrm{B}(\mathrm{x}) \Rightarrow \neg \mathrm{M}(\mathrm{x})\right)$.

For any x for which $\mathrm{B}(\mathrm{x})$ is true, we have $\mathrm{I}(\mathrm{x})$ true by 1., then $\mathrm{D}(\mathrm{x})$ true by 3., and finally $\neg \mathrm{M}(\mathrm{x})$ true by 2. Hence, the conclusion indeed follows from the premises.

Modelling in Predicate Logic

Example: Consider the following deduction:

1. All horses are animals.
2. *Therefore,* all horse-heads are animal heads.

Letting

- $\mathrm{Horse}(x) =$ "x is a horse"
- $\mathrm{Animal}(x) =$ "x is an animal" and
- $\mathrm{Head}(x, y) =$ "x is the head of y"

the statements above translate to:

1. $\forall x \Big(\mathrm{Horse}(x) \Rightarrow \mathrm{Animal}(x) \Big)$
2. $\forall x \Big(\exists y (\mathrm{Horse}(y) \wedge \mathrm{Head}(x, y)) \Rightarrow \exists y \Big(\mathrm{Animal}(y) \wedge \mathrm{Head}(x, y) \Big) \Big)$

Quantifiers and Propositional Connectives.

In the following, P(x) indicates that P contains x as a free variable.
P is also allowed to contain further free variables.

Seen: rules for negation

$$\neg\forall x P(x) \Leftrightarrow \exists x \neg P(x) \qquad \neg\exists x P(x) \Leftrightarrow \forall x \neg P(x)$$

Rules for "and":

$$\forall x.(P(x) \wedge Q(x)) \Leftrightarrow \forall x P(x) \wedge \forall x Q(x)$$

$$\exists x.(P(x) \wedge Q(x)) \Rightarrow \exists x P(x) \wedge \exists x Q(x)$$ This implication does not reverse.

$$\exists x.\exists y.(P(x) \wedge Q(y)) \Leftrightarrow \exists x P(x) \wedge \exists y Q(y)$$

Rules for "or":

$$\forall x.(P(x) \vee Q(x)) \Leftarrow \forall x P(x) \vee \forall x Q(x)$$

$$\exists x.(P(x) \vee Q(x)) \Leftrightarrow \exists x P(x) \vee \exists x Q(x)$$

Rules for "Implies":

$$\forall x\,(P \Rightarrow Q(x)) \Leftrightarrow P \Rightarrow \forall x Q(x) \qquad \exists x\,(P \Rightarrow Q(x)) \Leftrightarrow P \Rightarrow \exists x Q(x)$$

x does not appear free in P

$$(\exists x P(x) \Rightarrow Q) \Leftrightarrow \forall x (P(x) \Rightarrow Q) \qquad (\forall x P(x) \Rightarrow Q) \Leftrightarrow \exists x (P(x) \Rightarrow Q)$$

x does not appear free in Q